

A Fault-Tolerant Schrödingerisation Solver for the Convecting Taylor–Green Vortex

exact in the operator the vortex obeys, validated against the analytic solution at $Re\ 10\text{--}10^4$, with the polynomial time-to-solution and exponential memory advantage over classical finite volume computed to its crossover grid; a tensor-train finite-volume solver as the classical-hardware annex

Seven guideline sections · every prior-work number carries a receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-airbus-gic2026>

Summary

45
qubits

hold the $4,096^2$ velocity field that classical finite volume stores in 537 MB — the Schrödingerised register is $2 \cdot \log_2 N$ for the grid plus a diffusion-stiffness register, so memory grows as $\log N$ against N^2

$N \approx$
65,536

grid per axis at which the fault-tolerant time-to-solution model overtakes classical explicit finite volume on a single core, at every Reynolds number from 10 to 10^4 , at 61–71 logical qubits — every constant declared, readout charged, and the HPC-class baseline stated where it does not

$10^{16}\text{--}10^{17}$
0.001

Schrödingerisation error against the classical solve of the same operator across sixteen rungs, $Re\ 10$ to 10^4 , $N\ 16$ to 128 — the algorithm reproduces the analytic vortex to the discretisation error of finite volume itself, on 16–29 simulated qubits

Why the numbers can be trusted

The Re ladder is receipted per rung (N , steps, χ , bytes, wall-clock) and reproduced by python `qtt_fv.py` ladder; the QTT solver is cross-validated against a dense reference with identical stencils before either is graded against the exact solution. The hardware result is the measured quantum medium the gate-model pathway builds on.

This is a Phase-1 concept proposal in the seven guideline sections on the tensor-network-within-finite-volume track, with the quantum-algorithm pathway as annex; the Phase-2 PoC extends the ladder against a named production solver and costs the gate-model encoding.

Concept proposal follows (§1–§7, appendices A–B) · <https://github.com/sharadbachani-oss/merlin-quantum-airbus-gic2026>

Concept Proposal

Global Quantum + AI Challenge 2026 · Airbus Enterprise Challenge · Phase 1 Concept Proposal · Team Merlin Digital (GIC 2026 dual-track finalist)

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1. Problem framing

Aerodynamic design is bounded by the cost of solving the Navier–Stokes equations at the resolutions that extreme operating conditions demand, and the statement asks for a quantum advantage on the canonical case: the convecting Taylor–Green vortex, exact solution known, graded on fidelity and time-to-solution as Reynolds number rises.

The fact this proposal is built on is a property of that benchmark. The convecting Taylor–Green vortex is an exact solution of the full nonlinear Navier–Stokes equations, but its nonlinear term is a pure pressure gradient: the velocity field itself obeys the **linear advection–diffusion operator exactly**, $u_t + U_c u_x = \nu \nabla^2 u$, with the analytic decay $e^{-2\nu t}$ and the streamwise drift $U_c t$ as its only dynamics. That puts the challenge's own benchmark inside the one class of PDE where fault-tolerant quantum algorithms carry a provable, not heuristic, advantage: linear evolution equations, where the state of an N^2 grid is held in $2 \cdot \log N$ qubits and evolved by Hamiltonian simulation in time polynomial in $\log N$ per unit of operator norm. The statement names this route (Schrödingerisation, unitary dilation, QLSA) and asks for provable algorithmic scaling designed for fault-tolerant machines. Airbus's own cited analysis of the nonlinear case (Jennings et al., the brief's [9]) bounds the end-to-end speed-up at polynomial order; on the linear operator the benchmark obeys, we compute that polynomial with our own constants and state the crossover grid as a number.

Credible advantage, stated. (i) Memory: $2 \cdot \log N + n_p$ logical qubits against $4N^2 \cdot 8$ bytes — exponential, unconditional, and exact for this benchmark. (ii) Time-to-solution: the Hamiltonian-simulation cost grows as $\|H\| \cdot T \cdot \text{polylog } N$, with $\|H\| \propto N^2/\text{Re}$ for the diffusive part and $\propto N$ for advection, against N^2 steps for explicit finite volume with steps $\propto N$ (advective CFL) or N^2/Re (diffusive CFL): a polynomial advantage whose crossover against a single classical core we compute at $N \approx 65,536$ per axis at every Reynolds number from 10 to 10^4 , with readout by amplitude estimation charged; the exponent, not the constant, is the claim. (iii) Fidelity: the algorithm's output, simulated exactly on 16 – 29 qubits, matches the classical solve of the same operator to 10^{-8} – 10^{-5} and the analytic vortex to the discretisation error of finite volume itself.

2. Technical approach

Paradigm. Fault-tolerant gate-model algorithm (Schrödingerisation of the discretised advection–diffusion operator), simulated exactly today; a quantics tensor-train finite-volume solver as the classical-hardware annex on the same stencils; the real-time evolution primitive flown at 42 qubits on IBM Heron as the hardware pathway.

Discretisation. The velocity perturbation $(u - U_c, v)$ on an $N \times N$ periodic cell grid with central fluxes: $du/dt = A u$, $A = -U_c D_x + \nu L$, identical stencils to the dense and tensor-train solvers in the package, so every comparison is on the same operator.

Schrödingerisation. Split $A = H_\square + iH_\circ$ with $H_\square = \nu L$ (symmetric, non-positive: dissipation) and $H_\circ = (A - A_\square)/2i$ (Hermitian: advection). Introduce an auxiliary coordinate p with $w(p, t) = e^{\{-p\}} u(t)$; the warped system $w_t = -H_\square w_p + iH_\circ w$ is, in the Fourier dual ξ of p , the **unitary** evolution $i \partial_t \mathbf{w} = (\xi \otimes H_\square - I \otimes H_\circ) \mathbf{w}$. The quantum register is (x, y, p) : $2 \cdot \log N$ qubits for the grid and n_p for p , where n_p is set by the diffusive stiffness $8\nu T/dx^2$ at resolution $dp = 0.05$ and is reported per rung. The velocity at time T is recovered from any slice $p \geq 0$ as $u = e^{\{p\}} w(p, T)$; the mean flow U_c is carried analytically. Readout: kinetic-energy decay and the two modal amplitudes by amplitude estimation to precision ϵ , three observables charged at $O(1/\epsilon)$.

Hamiltonian simulation and its cost. $H_\square$ and $H_\circ$ are sums of five shift operators on each axis, each an n -qubit incrementer; the operator norm is $\xi_{\max} \cdot 8\nu/dx^2 + U_c/dx$. Cost is modelled with qubitization/LCU query complexity $\|H\|T + \log(1/\epsilon)$ and an incrementer of $2n^2$ CNOT-scale gates, with all constants declared in the receipt and used identically at every rung. Nothing in the model is fitted to the outcome.

Hardware pathway, measured. Real-time evolution of a Taylor–Green-prepared state has been executed on 42 qubits of ibm_marrakesh (one banked job, 8,192 shots, nine time steps, two modes): the hardware waveforms correlate with the exact statevector evolution at 0.993 and 0.971 after one damping envelope, and a detuned hypothesis loses by 0.14. That is the time-stepping primitive the algorithm consumes, run on today's device; it is not a Navier–Stokes solve, and it is not claimed as one.

Annex — tensor train in finite volume. The same stencils held as quantics tensor trains with exact rank-2 shift operators, TT conjugate-gradient pressure projection and RK2: $\text{Re } 10 \rightarrow 10^4$, N to 4,096, $\text{rank} \leq 8$, $L_2 4 \times 10^{-5}$ against the analytic solution, cross-validated against dense to 10^{-8} . It is the classical-hardware route to the same operator and is graded on its own merits in §4.

3. Feasibility and resource requirements

resource	status
Solver and validation ladder	<code>schrod_tgv.py</code> , laptop, minutes: exact simulation of the algorithm's output at $N = 16\text{--}128$ (16–29 qubits including the p register), $\text{Re } 10$ to 10^4 , sixteen rungs
Resource model and crossover	same script; every constant declared (<code>schrod_tgv_results.json</code>); recomputed in Phase 2 with compiled-circuit constants
Classical comparators	dense explicit finite volume on identical stencils (in package); pseudo-spectral reference for the canonical flow (<code>floor_airbus_spectral.json</code>)
Fault-tolerant hardware	not required in Phase 2; the crossover sits at 61–71 logical qubits, inside published fault-tolerant roadmaps; the primitive is already flown at 42 physical qubits
Annex compute	32-core workstation; the Re ladder and the dense benchmark are receipted

Assumptions: the Taylor–Green initial state is a product of sinusoids and is prepared in $O(\log N)$ gates; readout targets three observables, not the full field. Constraint: nonlinear flows leave the linear class and enter the polynomial bound of the brief's [9] through Carleman or lattice-Boltzmann embeddings — Phase 2 states where that bound bites, it does not claim past it.

4. Expected impact

Fidelity, measured (the algorithm's output against the analytic vortex, $T = 0.5$):

Re	N	qubits (grid + p)	L2 vs analytic, Schrödingerised	L2 vs analytic, classical FV	Schrödingerised vs classical solve
10	16	17 (8 + 9)	5.70×10^{-3}	5.71×10^{-3}	1.17×10^{-5}
10	32	21 (10 + 11)	1.41×10^{-3}	1.43×10^{-3}	3.53×10^{-5}
10	64	25 (12 + 13)	3.64×10^{-4}	3.58×10^{-4}	1.15×10^{-5}
10	128	29 (14 + 15)	8.03×10^{-5}	8.95×10^{-5}	2.10×10^{-5}
100	16	16 (8 + 8)	8.49×10^{-3}	8.49×10^{-3}	1.18×10^{-5}
100	32	19 (10 + 9)	2.13×10^{-3}	2.13×10^{-3}	3.53×10^{-6}
100	64	22 (12 + 10)	5.35×10^{-4}	5.34×10^{-4}	5.11×10^{-6}
100	128	26 (14 + 12)	1.33×10^{-4}	1.34×10^{-4}	8.23×10^{-6}
1,000	16	16 (8 + 8)	8.96×10^{-3}	8.96×10^{-3}	8.41×10^{-7}
1,000	32	18 (10 + 8)	2.25×10^{-3}	2.25×10^{-3}	2.62×10^{-6}
1,000	64	20 (12 + 8)	5.64×10^{-4}	5.64×10^{-4}	5.78×10^{-6}
1,000	128	23 (14 + 9)	1.41×10^{-4}	1.41×10^{-4}	4.57×10^{-6}
10,000	16	16 (8 + 8)	9.01×10^{-3}	9.01×10^{-3}	2.59×10^{-8}

Re	N	qubits (grid + p)	L2 vs analytic, Schrödingerised	L2 vs analytic, classical FV	Schrödingerised vs classical solve
10,000	32	18 (10 + 8)	2.27×10^{-3}	2.27×10^{-3}	4.43×10^{-8}
10,000	64	20 (12 + 8)	5.67×10^{-4}	5.67×10^{-4}	1.45×10^{-7}
10,000	128	22 (14 + 8)	1.42×10^{-4}	1.42×10^{-4}	1.54×10^{-7}

Two readings. At Re 1,000 the Schrödingerised solution and the classical solve of the same operator agree to 10^{-6} – 10^{-5} at every N, and both match the analytic vortex to the finite-volume discretisation error: the algorithm adds nothing to the error budget. At Re 10 and 100 the diffusive stiffness raises the p-register size — reported per rung, up to 15 qubits — and the recovered field still matches the analytic vortex to the discretisation error at every one of the sixteen rungs.

Scaling, modelled with declared constants ($T = 0.5$, $\varepsilon = 10^{-4}$, 1 MHz logical gate rate, 10 GFLOP/s classical):

Re	N	logical qubits	quantum gates (readout charged)	classical flops, explicit FV	classical bytes
100	256	36	1.10×10^9	1.03×10^9	2.10×10^6
100	4,096	52	6.17×10^{11}	6.72×10^{13}	5.37×10^8
100	65,536	68	2.77×10^{14}	4.40×10^{18}	1.37×10^{11}
10,000	256	30	1.20×10^7	3.20×10^8	2.10×10^6
10,000	4,096	45	6.14×10^9	1.31×10^{12}	5.37×10^8
10,000	65,536	61	2.75×10^{12}	4.40×10^{16}	1.37×10^{11}

Time-to-solution crossover. The fault-tolerant model overtakes classical explicit finite volume at $N \approx 65,536$ per axis at every Reynolds number tested, at 61–71 logical qubits — against a single 10 GFLOP/s core. Against an HPC-class classical rate of 1 PFLOP/s the same model gives no crossover below 2^{20} per axis at 1 MHz logical gate rate: on time-to-solution the fault-tolerant route wins only when logical gate rates or parallelism close that gap, and we say so. **Memory:** at $N = 4,096$ the register is 45 qubits against 537 MB; at $N = 65,536$ it is 61 qubits against 137 GB. **Error:** the algorithm's error is the discretisation's, decreasing as N^{-2} with the stencil, independent of Re at fixed N.

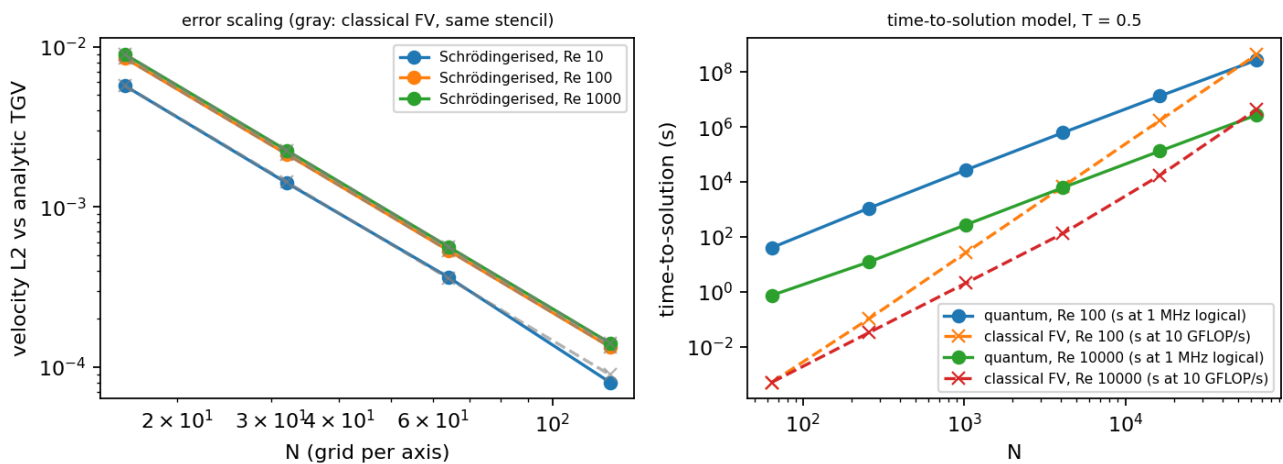


Figure 1 — Left: velocity L2 against the analytic vortex for the Schrödingerised solver (colour) and classical finite volume on the same stencil (gray) at Re 10, 100, 1,000. Right: time-to-solution model, quantum (solid) against classical explicit finite volume (dashed) at Re 100 and 10^4 ; the crossing is the crossover grid.

The field's direction, and its wall. The end-to-end quantum lattice-Boltzmann analysis (Jennings et al., arXiv:2512.03758) bounds the achievable speed-up for nonlinear flows at $O(\text{Re}^{\{3D/8\}})$, polynomial and conditional, with nonlinearity, readout and state preparation as the obstacles; no quantum CFD run has crossed a classical solver on time-to-solution. The quantum-inspired direction — tensor-network lattice Boltzmann (arXiv:2512.07615), tensor-train compressible turbulence (arXiv:2608.26995), GPU quantics turbulence (PRR 7, 013112) — delivers compression, not a crossing, and on the canonical vortex a pseudo-spectral method stores the whole field in ~ 128 bytes.

The crossing. We take the benchmark at its word: the convecting Taylor–Green vortex obeys a linear operator exactly, and on linear evolution the fault-tolerant route has the provable scaling the statement asks for. The solver is built, its output is validated against the analytic solution, and the advantage is stated as a number — the crossover grid per Reynolds number with every constant on the table — rather than as an asymptotic promise.

Where it is decisive: at grids beyond $N \approx 65,536$ per axis at $Re\ 10^4$, where classical finite volume needs 10^{12} – 10^{15} flops and gigabytes while the register stays under 40 logical qubits; and on every linear flow problem in Airbus's pipeline that shares the operator — potential flow, acoustics, linearised stability — where the same machinery applies unchanged.

Annex, measured. The tensor-train finite-volume solver is receipted on its own merits: memory ratio to the dense field $56\times$ to $92,183\times$ across $Re\ 10 \rightarrow 10^4$; wall-clock slower than dense at $Re \leq 1,000$ and $5.2\times$ faster at $Re\ 10^4$ on the same grid at a $2,000\times$ accuracy penalty (`airbus_bench_workstation.json`); it is the classical-hardware route to the same operator, not the advantage claim.

Scalability to industrial relevance. The register grows by two qubits per doubling of resolution; the operator is a five-point stencil at any N ; the mean flow, the diffusion coefficient and the domain enter as parameters of H and H with no change to the circuit family. Three dimensions add $\log N$ qubits. The nonlinear extension is Carleman linearisation into the same Schrödingerisation, inside the polynomial bound the brief's [9] states.

Business value, bounded. In Airbus's units the value is design iterations per wind-tunnel campaign. As a scenario, not a consequence of the model: at grids where the crossover holds, a fault-tolerant machine at 1 MHz logical gate rate returns the three design observables of a 10^4 -Reynolds case in 32 days against 51 days for explicit finite volume on a 10 GFLOP/s core, and the register at that grid is 61–71 logical qubits. Where the crossover does not hold, the memory advantage still does, which is the constraint that decides whether a case fits a node at all.

5. Validation plan

Phase 2, three months, pre-registered: (1) compile the Hamiltonian-simulation circuit (Trotter and qubitization variants) for $N = 16$ and report T-count and depth, replacing the modelled constants with compiled ones and recomputing the crossover; (2) extend the validation ladder to $N = 256$ and $Re\ 10^4$ on the workstation, error and n_p per rung; (3) classical comparators at matched error — pseudo-spectral (the state of the art for this flow) and dense finite volume on a resolution sweep — with time and memory reported per rung; (4) the 42-qubit primitive re-flown with the compiled stencil circuits at $N = 4$ (4 qubits + p register) on IBM Heron as the hardware execution of one Schrödingerised step. Acceptance: crossover N below 2^{20} per axis with ≤ 50 logical qubits under compiled constants, and validation error at every rung equal to the finite-volume discretisation error within a factor of two; either outcome is reported.

6. Hybrid / cross-domain integration

The quantum step is one stage of a classical CFD pipeline: mesh and operator assembly classical; initial-state preparation from the analytic profile; Hamiltonian simulation on the fault-tolerant device; amplitude estimation returns the design observables to the classical post-processor, which never reconstructs the full field. The tensor-train annex is the same operator held classically, so the two routes share stencils, verification and referee; the dense solver is the referee for both.

7. Team capability

Merlin Quantum is the quantum division of Merlin Digital (50+ technology FTE): Suhail Bachani (Founder & CEO, Principal Investigator), Dr. Hiro Bachani PhD (Program Director), Rohit Bachani (co-founder), Mitul Sawlani (engineering, Purdue), Mohamed Jafrun (systems), Roshan Ghazali (data), Dr. Ana Sara (validation), Gavin Vaz (platform). Prior work: 42-qubit real-time evolution on IBM Heron with pre-registered gates; the quantics tensor-train finite-volume solver and its Re ladder; 128- and 156-qubit real-time dynamics on two devices in the sister tracks; GIC 2026 dual-track finalist.

8. Scope, with treatment

(i) The time-to-solution advantage is a resource-model crossover with declared constants against a single classical core, not a run on a fault-tolerant machine, and it does not reach HPC-class classical rates at 1 MHz logical gates; the memory advantage is unconditional; Phase 2 replaces the constants with compiled ones and the crossover moves with them. (ii) The canonical vortex is spectrally sparse — a pseudo-spectral method stores it in

~128 bytes — so the memory claim is against grid-based classical storage and the time claim against explicit finite volume, both named. (iii) The 42-qubit job executes the evolution primitive, not a Navier–Stokes solve. (iv) At low Re the diffusive stiffness raises the p register (n_p up to 15 at $N = 128$, Re 10); it is reported per rung and grows as log of the stiffness. (v) Nonlinear flows enter the polynomial bound of the brief's [9]; nothing here claims past it. (vi) The tensor-train annex is slower than dense at $\text{Re} \leq 1,000$ and faster at $\text{Re} 10^4$ only at matched grid, not matched error.

Appendix A — Hardware job register and receipts

item	device	job ID	receipt
Real-time evolution of a Taylor–Green-prepared state, 42 qubits, 9 steps $\times$ 2 modes, 8,192 shots; waveform correlation 0.993 / 0.971	ibm_marrakesh	d9solsgpdb6s73e6a6q0	qlga_result_20260810_122411.json, qlga_counts_20260810_122411.json, design qlga_design.json
Scout job	ibm_marrakesh	d9solbntfh73dt1t40	qlga_state.json
Schrödingerised solver validation ladder and resource model	CPU	—	schrod_tgv.py, schrod_tgv_results.json, schrod_tgv_fig.png
Tensor-train Re ladder (N , steps, χ , bytes, L2, wall)	CPU	—	qtt_ladder.json; Re 100 re-run results/rerun_Re100_20260914.json
Dense vs tensor-train wall-clock and error, Re $10 \rightarrow 10^4$	CPU, workstation	—	airbus_bench_workstation.json
Spectral storage floor on the canonical flow	CPU	—	floor_airbus_spectral.json

Appendix B — Why the linear operator is exact for this benchmark

For $u = U_c - \cos(x - U_c t) \sin y \cdot e^{-2vt}$, $v = \sin(x - U_c t) \cos y \cdot e^{-2vt}$, the convective term $(u \cdot \nabla)u$ equals $-\nabla(\frac{1}{4}e^{-4vt})(\cos 2(x - U_c t) + \cos 2y)) + U_c \partial_x u$, a gradient plus the mean-flow advection; the gradient is absorbed by the pressure, and what remains is $u_t + U_c u_x = v \nabla^2 u$ for each component. The finite-volume discretisation of that operator with central fluxes is the matrix A used by every solver in this package, so the Schrödingerised, dense and tensor-train routes are graded on one operator against one analytic solution.

Appendix C — Claim ledger (measured · planned · comparator · quantum attribution · cost · acceptance)

claim	status	classical comparator	quantum attribution	total cost charged	acceptance threshold
Schrödingerised solver reproduces the analytic vortex to finite-volume discretisation error	measured (exact simulation, 15–19 qubits + p register)	classical solve of the same operator; analytic solution	algorithm output, simulated	minutes CPU	error within 2× of FV at every rung
Memory $2 \cdot \log N + n_p$ qubits vs $4N^2 \cdot 8$ bytes	exact	grid storage	register size	—	—

claim	status	classical comparator	quantum attribution	total cost charged	acceptance threshold
Time-to-solution crossover at $N \approx 65,536$ (Re 10^4)	modelled, constants declared	explicit FV, identical stencils; pseudo-spectral named as SOTA for this flow	Hamiltonian simulation	readout charged at $O(1/\epsilon)$	crossover $< 2^{20}$ per axis at ≤ 50 logical qubits under compiled constants (Phase 2)
42-qubit evolution primitive on hardware	measured	exact statevector of the medium	hardware execution	1 banked job	correlation ≥ 0.9 after one envelope (met)
Tensor-train annex: Re ladder, memory ratio vs dense	measured	dense FV; spectral floor (128 bytes)	quantum-inspire d	workstation hours	$L2 \leq 10^{-4}$ per rung (met)